

```
/* CMP-135 SDTM DM Creation
Domain: DM*/;

options validvarname=uppercase nofmterr dlcreatedir;

libname raw "/home/u64124548/sasuser.v94/SDTM - Creation/crfdatav1";
libname sdtm "/home/u64124548/sasuser.v94/SDTM - Creation/output_sdtm";

libname valxpt xport
"/home/u64124548/sasuser.v94/SDTM - Creation/SDTM Data For Validation V1/dm.xpt";
```

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```
%macro make_date(invar=, outdt=);
    &outdt = .;

    if vtype(&invar) = "N" then do;
        if not missing(&invar) then do;
            if abs(&invar) > 50000 then &outdt = datepart(&invar);
            else &outdt = &invar;
        end;
    end;
else do;
    _dtm = input(strip(&invar), anydtdtm.);
    if not missing(_dtm) then &outdt = datepart(_dtm);

    if missing(&outdt) then
        &outdt = input(strip(&invar), anydtdte.);
end;
%mend;
```

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```
%macro make_dtm(datevar=, timevar=, outdt=, outdtm=);
    %make_date(invar=&datevar, outdt=&outdt);

    _tm = .;

    if vtype(&timevar) = "N" then _tm = &timevar;
    else _tm = input(strip(&timevar), time.);

    if not missing(&outdt) and not missing(_tm) then
        &outdtm = dhms(&outdt, hour(_tm), minute(_tm), second(_tm));
    else if not missing(&outdt) then
        &outdtm = dhms(&outdt, 0, 0, 0);
%mend;
```

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```
data ex_prep;
    set raw.ex;

    %make_dtm(datevar=exstdtn, timevar=exsttm, outdt=exdt, outdtm=exdtm);

    format exdt date9. exdtm datetime20.;
run;
```

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```
proc sql;
    create table ex_dates as
    select
```

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    subject,  
    min(exdt)  as first_dose_dt  format=date9.,  
    max(exdt)  as last_dose_dt   format=date9.,  
    min(exdtm) as first_dose_dtm format=datetime20.,  
    max(exdtm) as last_dose_dtm  format=datetime20.  
from ex_prep  
where not missing(exdt)  
group by subject;  
quit;
```

---

```
data ds_prep;  
  set raw.ds;  
  
  %make_date(invar=dsstdat, outdt=dsdt);  
  
  format dsdt date9.;  
run;
```

---

```
proc sql;  
  create table ds_dates as  
  select  
    subject,  
    max(dsdt) as eos_dt format=date9.  
  from ds_prep  
  where not missing(dsdt)  
  group by subject;  
quit;
```

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```
data enr_dates;  
  set raw.enr;  
  
  %make_date(invar=cnstdtn, outdt=consent_dt);  
  
  format consent_dt date9.;  
  
  keep subject consent_dt enrgrp;  
run;
```

---

```
data inv_prep;  
  set raw.inv;  
  
  length siteid $4 invnam $13;  
  
  siteid = strip(siteid);  
  invnam = catx(" ", invfname, invlname);  
  
  keep siteid invid invnam country;  
run;
```

---

```
data dov_prep;  
  set raw.dov;  
  
  %make_date(invar=visdtn, outdt=dmdt);
```

```

folder_u = upcase(strip(foldername));
inst_u   = upcase(strip(instancename));

format dmdt date9.;
run;

```

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```

proc sql;
create table dov_screen as
select
    subject,
    min(dmdt) as dmdt format=date9.
from dov_prep
where index(folder_u, "SCREEN") > 0
    or index(inst_u, "SCREEN") > 0
group by subject;
quit;

```

---

```

proc sql;
create table dm_merge as
select
    a.*,
    b.first_dose_dt,
    b.last_dose_dt,
    b.first_dose_dtm,
    b.last_dose_dtm,
    c.eos_dt,
    d.consent_dt,
    d.enrgp,
    f.invid,
    f.invnam,
    f.country,
    g.dmdt
from raw.dm as a
left join ex_dates   as b on a.subject = b.subject
left join ds_dates   as c on a.subject = c.subject
left join enr_dates  as d on a.subject = d.subject
left join inv_prep   as f on scan(a.subject, 1, "-") = f.siteid
left join dov_screen as g on a.subject = g.subject;
quit;

```

---

```

data sdtm.dm;
retain
    STUDYID DOMAIN SUBJID RFSTDTC RFENDTC RFXSTDTC RFXENDTC
    RFICDTC RFPENDTC DTHDTC DTHFL SITEID INVID INVNAM BRTHDTC
    AGE AGEU SEX RACE ETHNIC ARM ACTARM COUNTRY DMDTC DMDY
    USUBJID ACTARMCD ARMCD;

length
    STUDYID   $6
    DOMAIN    $2
    SUBJID    $9
    RFSTDTC   $10
    RFENDTC   $10
    RFXSTDTC  $16

```

```

RFXENDTC  $16
RFICDTC    $10
RFPENDTC   $10
DTHDTC     $1
DTHFL      $1
SITEID     $4
INVID      $4
INVNAM     $13
BRTHDTC    $10
AGEU       $5
SEX        $1
RACE       $5
ETHNIC     $22
ARM        $7
ACTARM     $7
COUNTRY    $3
DMDTC      $10
USUBJID    $40
ACTARMCD   $8
ARMCD      $8;

```

```
set dm_merge;
```

```
label
```

```

STUDYID = "Study Identifier"
DOMAIN  = "Domain Abbreviation"
SUBJID  = "Subject Identifier for the Study"
RFSTDTC = "Subject Reference Start Date/Time"
RFENDTC = "Subject Reference End Date/Time"
RFXSTDTC = "Date/Time of First Study Treatment"
RFXENDTC = "Date/Time of Last Study Treatment"
RFICDTC = "Date/Time of Informed Consent"
RFPENDTC = "Date/Time of End of Participation"
DTHDTC = "Date/Time of Death"
DTHFL = "Subject Death Flag"
SITEID = "Study Site Identifier"
INVID = "Investigator Identifier"
INVNAM = "Investigator Name"
BRTHDTC = "Date/Time of Birth"
AGE = "Age"
AGEU = "Age Units"
SEX = "Sex"
RACE = "Race"
ETHNIC = "Ethnicity"
ARMCD = "Planned Arm Code"
ARM = "Description of Planned Arm"
ACTARMCD = "Actual Arm Code"
ACTARM = "Description of Actual Arm"
COUNTRY = "Country"
DMDTC = "Date/Time of Collection"
DMDY = "Study Day of Collection";

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```
/* USUBJID = "Unique Subject Identifier"; */
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```
STUDYID = "CMP135";
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```
DOMAIN = "DM";
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```
SUBJID = strip(subject);
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```
USUBJID = catx("-", STUDYID, SUBJID);

RFSTDTC = put(first_dose_dt, yymmdd10.);
RFENDTC = put(eos_dt, yymmdd10.);
RFPENDTC = RFENDTC;

if not missing(first_dose_dtm) then
    RFXSTDTC = cats(
        put(datepart(first_dose_dtm), yymmdd10.),
        "T",
        put(timepart(first_dose_dtm), tod5.)
    );

if not missing(last_dose_dtm) then do;
    if timepart(last_dose_dtm) = 0 then
        RFXENDTC = put(datepart(last_dose_dtm), yymmdd10.);
    else
        RFXENDTC = cats(
            put(datepart(last_dose_dtm), yymmdd10.),
            "T",
            put(timepart(last_dose_dtm), tod5.)
        );
end;

RFICDTC = put(consent_dt, yymmdd10.);

DTHDTC = "";
DTHFL = "";

SITEID = scan(strip(subject), 1, "-");
INVID = strip(invid);
INVNAM = strip(invnam);

%make_date(invar=brthdtn, outdt=brthdt);

BRTHDTC = put(brthdt, yymmdd10.);

if not missing(brthdt) and not missing(consent_dt) then
    AGE = int(yrdif(brthdt, consent_dt, "ACTUAL"));
else AGE = agecalc;

AGEU = "YEARS";

SEX = strip(sex_cod);
RACE = upcase(strip(race));
ETHNIC = upcase(strip(ethnic));

ARMCD = "CMP135_5";
ARM = "Group 1";
ACTARMCD = "CMP135_5";
ACTARM = "Group 1";

COUNTRY = strip(country);

DMDTC = put(dmdt, yymmdd10.);

if not missing(dmdt) and not missing(first_dose_dt) then do;
```

```
    if dmdt >= first_dose_dt then DMDY = dmdt - first_dose_dt + 1;
    else DMDY = dmdt - first_dose_dt;
end;

keep
    STUDYID DOMAIN SUBJID RFSTDTC RFENDTC RFXSTDTC RFXENDTC
    RFICDTC RFPENDTC DTHDTC DTHFL SITEID INVID INVNAM BRTHDTC
    AGE AGEU SEX RACE ETHNIC ARM ACTARM COUNTRY DMDTC DMDY
    USUBJID ACTARMCD ARMCD;

run;

-----
proc sort data=sdm.dm out=dm_created;
    by STUDYID USUBJID;
run;

-----
proc sort data=valxpt.dm out=dm_validate;
    by STUDYID USUBJID;
run;

ods rtf file="/home/u64124548/sasuser.v94/SDTM - Creation/output_sdtm/DM_PROC_COMPARE.rtf";

title "PROC COMPARE: SDTM DM Created vs Validation DM";

-----
proc compare base=dm_validate compare=dm_created listall criterion=0.0000001;
    id STUDYID USUBJID;
```